5.1. RESONANCE

Resonance is that condition in any A.C. circuit at a particular frequency when the applied voltage and resultant current are in same phase. So at resonance the power factor of the a.c. circuit is unity and it behaves like a pure resistive circuit.

5.2. SERIES RESONANCE

In series RLC circuit at a particular frequency, when inductive reactance $(X_L = \omega L)$ becomes equal to the capacitive reactance $(X_C = 1/\omega C)$, then the circuit is said to be in resonance, *i.e.*, at resonance.

$$X_L = X_C \qquad ...(5.1)$$

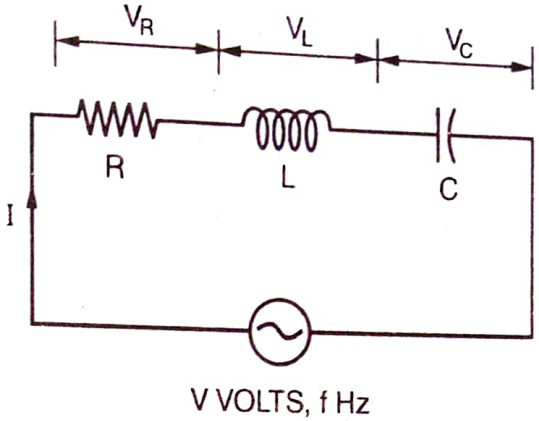


Fig. 5.1. Series Resonance Circuit.

In the series RLC circuit of Fig. 5.1, the circuit current I is given by

$$I = \frac{V}{Z} \; A$$

where Z represents the equivalent impedance of the circuit.

$$Z = R + j\omega L + \frac{1}{j\,\omega C} = R + j\omega L - \frac{j}{\omega C}$$

or, $$Z = R + j\,(X_L - X_C)$$

where $$X_L = \omega\, L \text{ and } X_C = \frac{1}{\omega C}$$

where $(X_L - X_C)$ is X, the net reactance

Thus, $$I = \frac{V}{R + j\,(X_L - X_C)} = \frac{V}{R + j\,X} \text{ Amp.}$$

and voltage drop across resistance R is

$V_R = IR$ (in phase with I)

voltage drop across inductance L is

$V_L = I\,X_L$ (leading I by 90°)

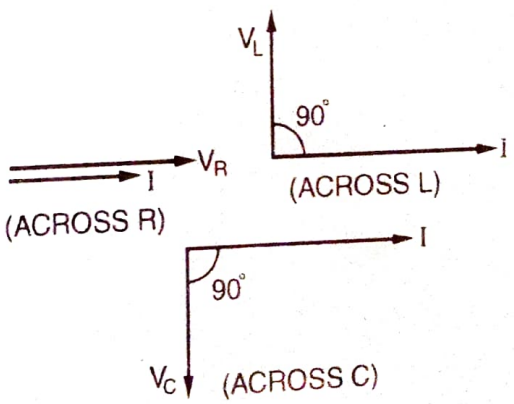


Fig. 5.2. Voltage and Current Phasors.

voltage drop across capacitance C is

$$V_C = I X_C \text{ (lagging } I \text{ by } 90^\circ).$$

The phasor diagram of the variables of the entire circuit has been shown in Fig. 5.3.

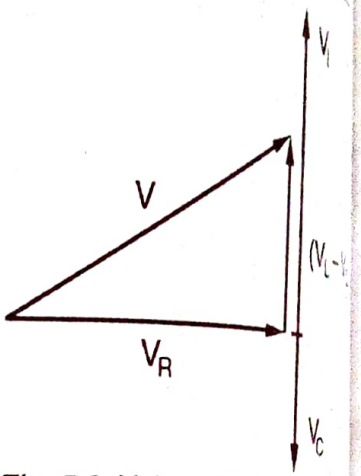


Fig. 5.3. Voltage Vector Diagram RLC Series Circuit

Resonance conditions is obtained when $X_L = X_C$ in the series circuit [this is obtained by either decreasing the supply frequency when X_L would decrease and X_C would increase or by increasing the supply frequency making X_I to increase and X_C to decrease]

When $X_L = X_C$, $Z = R + j0 = R$

Also, $I_0 = \dfrac{V}{Z} = \dfrac{V}{R}$ Amp [I_0 = current at resonance]

The p.f (power factor) of the circuit becomes $\cos \phi = \dfrac{R}{Z} = \dfrac{R}{R} = 1$ ...(5.2

The expression of frequency of resonance can be obtained as follows :

Let ω_0 or f_0 be the frequency at which $X_L = X_C$

i.e., $\omega_0 L = \dfrac{1}{\omega_0 C}$ or, $\omega_0^2 = \dfrac{1}{LC}$

i.e., $\omega_0 = \dfrac{1}{\sqrt{LC}}$ rad/sec ...(5.3

i.e., $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$ Hz

Voltage vector diagram at resonance is shown in Fig. 5.4.



frequency is

$$Y = \frac{1}{Z} \qquad ...(5.5)$$

5.2.4. Resonance curve. Fig. 5.8 shows the variation of current I with frequency. Current is maximum at resonance frequency ω_0. At rms value of current $I_0/\sqrt{2}$, power becomes half of its maximum value. So the points corresponding to $I_0/\sqrt{2}$ on the resonance curve, are known as the half power points. Frequencies ω_1 and ω_2 corresponding to these two points are known as the lower and upper half power frequency.

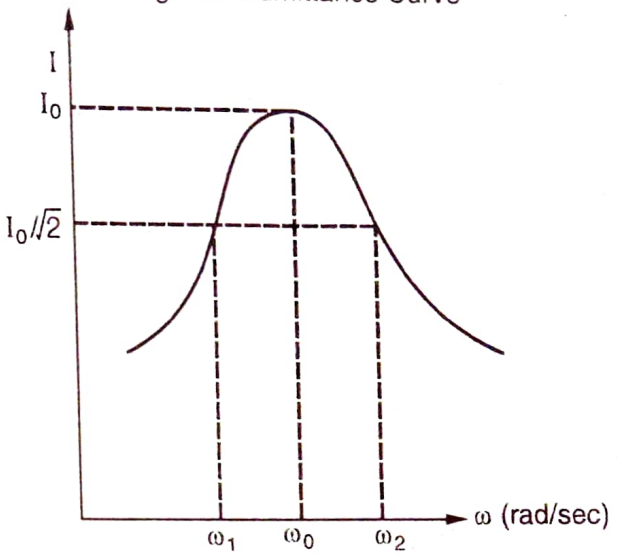


Fig. 5.8. Resonance Curve.

5.2.5. Band width. The distance between upper and lower half power frequencies, measured in hertz or in rad/sec, is called the band width (BW).

$$BW = \omega_2 - \omega_1 \text{ in rad/sec} \qquad ...(5.6)$$

or
$$BW = f_2 - f_1 \text{ in } HZ \qquad ...(5.7)$$

The resonant frequency ω_0 is geometric mean of ω_1 and ω_0

i.e., $\omega_0 = \sqrt{\omega_1 \omega_2}$ and $f_0 = \sqrt{f_1 f_2}$ ...(5.8)

In resonant circuit $\omega_2 - \omega_1 = R/L$ ...(5.9)

5.2.6. Expression of Half Power Frequencies in RLC Series Resonating Circuits

From equation 5.8, $\omega_2 \cdot \omega_1 = \omega_0^2$ ...(5.10)

$\Rightarrow$ $\omega_0 = \sqrt{\omega_1 \omega_2}$

Also, at resonance, $$\omega_0 = \frac{1}{\sqrt{LC}} \quad \text{...(5.11)}$$

Thus, the resonant frequency is the geometric mean of the two half power frequencies.

Again, from equation 5.10, $\omega_1 \omega_2 = \frac{1}{LC}$ and from equation 5.9, $(\omega_2 - \omega_1) = \frac{R}{L}$; substituting the value of ω_2 from 5.10 in equation 5.9.

$$\frac{1}{\omega_1 LC} - \omega_1 = \frac{R}{L}$$

or, $$\frac{1 - \omega_1^2 LC}{\omega_1 LC} = \frac{R}{L}$$

or, $$1 - \omega_1^2 LC = \omega_1 RC$$

or, $$\omega_1^2 LC + \omega_1 RC - 1 = 0$$

or, $\therefore$ $$\omega_1 = \frac{-RC \pm \sqrt{R^2 C^2 + 4LC}}{2LC} = \frac{-R \pm \sqrt{R^2 + 4\frac{L}{C}}}{2L} \quad \text{...(5.12)}$$

Taking the + ve sign of the radical,

$$\omega_1 = \frac{-R + \sqrt{R^2 + 4\frac{L}{C}}}{2L} = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \quad \text{...(5.13)}$$

Similarly, $$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \quad \text{...(5.14)}$$

5.2.7. Properties of Resonance of RLC Series Circuit.

1. The applied voltage and the resulting current are in phase which also mean that the p.f. of the RLC series resonant circuit is unity.
2. The net reactance is zero at resonance and the impedance does have the resistive part only.
3. The current in the circuit is maximum and is (V/R) Amp. [Since at resonance, the line current in the series LCR circuits is maximum hence it is called *acceptor circuit*].
4. At resonance, the circuit has got minimum impedance and maximum admittance.
5. Frequency of resonance is given by

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz.}$$

5.3. PARALLEL RESONANCE

Now we study the resonance condition in any type of parallel connection. At resonance in parallel circuit :

Imaginary part of admittance = 0 ...(5.15)

Fig. 5.9 represents a parallel resonating circuit where a coil is connected in parallel with a capacitor C and the combination is connected across an *a.c.* voltage source of variable frequency. Fig. 5.10 represents the vector diagram of the given circuit.

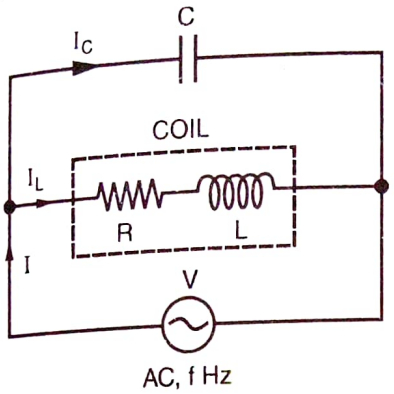


Fig. 5.9. A Typical parallel Resonating Circuit.

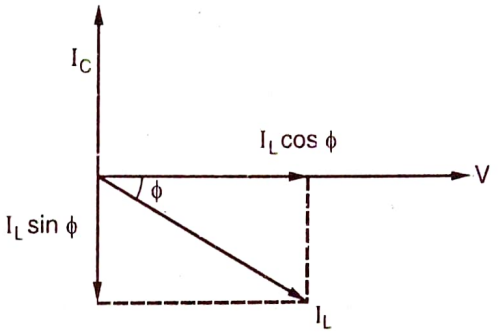


Fig. 5.10. Vector Diagram of the Parallel A.C. Circuit.

Let I_C = the current through the capacitor,
I_L = the current through the coil,
I = vector sum of I_L and I_C i.e., the source current,
V = supply voltage,
V_R = drop across R,
V_L = drop across L,
V_C = drop across C,
ϕ = p.f. angle of the coil (i.e., the angle of lag of I_L than V,)
Z_L = coil impedance,
X_C = capacitive reactance).

Here $$I_C = \frac{V}{X_C} \qquad ...(5.16)$$

and $$I_L = \frac{V}{Z_L} = \frac{V}{\sqrt{R^2 + X_L^2}} = \frac{V}{\sqrt{R^2 + (\omega L)^2}}$$

$$\cos \phi = \frac{R}{Z_L}.$$

At resonance, the capacitive current must be equal to the inductive part of the coil current *i.e.*, the imaginary components of I_L and I_C must cancel each other at resonance.

i.e., $$I_C = I_L \sin \phi \qquad \text{or,} \qquad \frac{V}{X_C} = \frac{V}{Z_L} \times \frac{X_L}{Z_L}$$

where $$\sin \phi = \frac{X_L}{Z_L},$$

or, $Z_L^2 = X_C X_L$

Also, $Z_L^2 = \frac{1}{\omega_0 C} \times \omega_0 \cdot L = \frac{L}{C}$ *i.e.*, $Z_L = \sqrt{\frac{L}{C}}$ ...(5.17)

[ω_0 represents resonance frequency]

or, $\sqrt{R^2 + \omega_0^2 L^2} = \sqrt{\frac{L}{C}}$ or, $\omega_0^2 L^2 = \frac{L}{C} - R^2$

i.e., $\omega_0^2 = \frac{1}{LC} - \frac{R^2}{L^2}$

$\therefore$ $\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} = \frac{1}{L}\sqrt{\frac{L}{C} - R^2}$ ...(5.18)

i.e., $f_0 = \frac{1}{2\pi L}\sqrt{\frac{L}{C} - R^2}$ ...(5.19)

If the resistance of the coil be neglected,

$$f_0 = \frac{1}{2\pi L}\sqrt{\frac{L}{C}} = \frac{1}{2\pi\sqrt{LC}} \quad \text{...(5.20)}$$

[*i.e.*, the frequency of resonance of series circuit].

Again, at resonance, since the reactive components of I_L and I_C balances each other, the only remaining part of the current is $I_L \cos\phi$ $(= I)$, $I = I_L \cos\phi$

or, $\frac{V}{Z_\Omega} = \frac{V}{Z_L} \cdot \frac{R}{Z_L}$ [Z_Ω = equivalent impedance of parallel circuit.]

or, $Z_\Omega = \frac{Z_L^2}{R} = \frac{L/C}{R} = \frac{L}{CR}$ [$\because$ from equation 5.27, $Z_L = \sqrt{L/C}$] ...(5.21)

Then, the equivalent impedance of the parallel resonating circuit is $\frac{L}{CR}$ at resonance.

This impedance is called *dynamic resistance* of the parallel circuit. Normally R being loss, this impedance is *very high* at resonance and then the current is *much lower* in the parallel circuit. Then, this circuit is also called *rejector circuit.*

5.3.1 Different ... admittance (Y) ...

Resonance Curve. Since admittance is minimum or impedance is maximum at resonance frequency ω_0, so current I is also minimum at resonance. (Fig. 5.12).

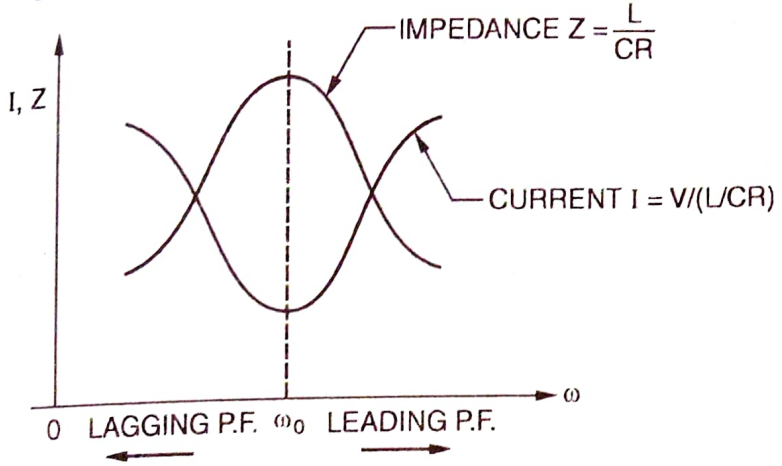


Fig. 5.12. Resonance Curve for Parallel Resonance.

5.3.2. Properties of Resonance in parallel RLC circuit.

1. Power factor is unity.
2. Current at resonance is $\left[\frac{V}{L/CR}\right]$ and is in phase with the applied voltage. The value of current at resonance is minimum. So the circuit is called the rejector circuit.
3. Net impedance at resonance of the parallel circuit is maximum and equal to $\frac{L}{CR}\Omega$.
4. The admittance is minimum and the net susceptance is zero at resonance.
5. The resonance frequency of this circuit is given by

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}.$$

Example 5.1. *Find the resonant frequency ω_0 (or f_0) for the given series RLC circuit.*

Also find the expression for resonant current.

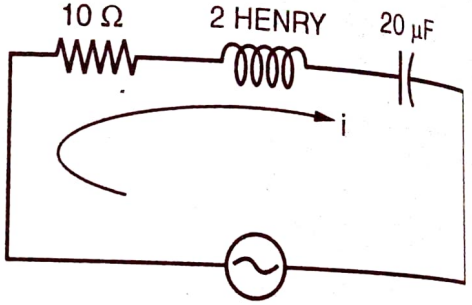


Fig. 5.13.

Solution. $R = 10\ \Omega$, $L = 2$ Henry, $C = 20$ microfarad

Under resonance :

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\therefore \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2 \times 20 \times 10^{-6}}} = 0.158 \times 10^3 \text{ rad/sec.}$$

and $$f_0 = \frac{1}{2\pi\sqrt{LC}} = 25.16 \text{ Hz}$$

Expression for current i is given by $i = \frac{v}{z}$ but at resonance $Z = R$ or $10\ \Omega$

so $$i = \frac{V_m \sin \omega t}{R} = \frac{50 \sin \omega t}{10} = 5 \sin \omega t \text{ Amp.}$$

is fulfilled.

Example 5.3. *A circuit having a resistance of 5 Ω, an inductance of 0.4 H and a variable capacitance in series, is connected across a 110V, 50 Hz supply. Calculate (i) The value of capacitance to give resonance, (ii) Current, (iii) Voltage across the inductance, (iv) Voltage across the capacitance, (v) Q-factor of the circuit.*

[MACT, Bhopal, 1999 ; BU, Bhopal, 1997]

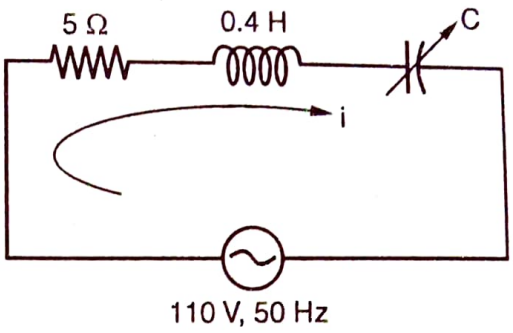


Fig. 5.15.

Solution.

Applied voltage, $V = 110\ V$

Resistance, $R = 5\ \Omega$

Inductance, $L = 0.4$ Hz

Resonant frequency $f_0 = 50$ Hz

As Resonant frequency $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$

(*i*) At resonant frequency capacitance

$$C = \frac{1}{4\pi^2 f_0^2 L} = \frac{1}{4\pi^2 \times (50)^2 \times 0.4}$$

$$= 2.53303 \times 10^{-5} = 25.3303\ \mu F$$

(*ii*) At resonant condition, $I = \dfrac{V}{R} = \dfrac{110}{5} = 22$ A

(*iii*) Voltage across inductance $IX_L = 22 \times 2\pi \times 50 \times 0.4$

$$= 2764.60 \text{ V}$$

(*iv*) Voltage across capacitance = Voltage across inductance

$$= 2764.60 \text{ V}$$

(*v*) Q-Factor $= \dfrac{\omega_0 L}{R} = \dfrac{2\pi f_0 L}{R} = \dfrac{2\pi \times 50 \times 0.4}{5} = 25.1327$

Example

Resistance of coil

Example 5.5. *A series circuit consists of 0.5 capacitor, a coil of inductance 0.32 H and resistance 40 Ω and a 20 Ω non-inductive resistor. Calculate the value of the resonant frequency of the circuit. When the circuit is connected to a 30V. AC supply, at this resonant frequency determine (i) The p.d across each of the 3 components, (ii) The resistance and inductance of the circuit (iii) The power absorbed.*

[BU, Bhopal, 1996, 1997]

Solution. Total Resistance, $R = 40\ \Omega + 20\ \Omega = 60\ \Omega$

Inductance, $L = 0.32$ H

Capacitance, $C = 0.5\ \mu\text{F} = 0.5 \times 10^{-6}$ F

Voltage applied $= 30$ V

Resonant frequency $$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.32 \times 0.5 \times 10^{-6}}} = 397.887 \text{ Hz}$$

Current at resonant condition $= \frac{V}{R} = \frac{30}{60} = 0.5$ A

Impedance of coil at resonant frequency

$$Z_L = \sqrt{R_L^2 + X_L^2}$$

$$= \sqrt{(40)^2 + (2\pi \times 397.887 \times 0.32)^2}$$

$$= 800.9987\ \Omega$$

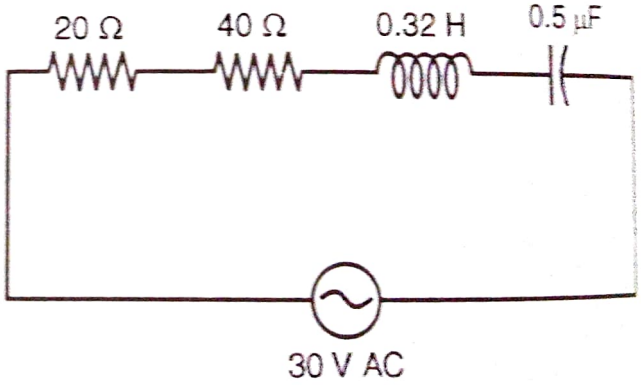


Fig. 5.17.

(*i*) (*a*) p.d. across coil $= Z_L \times I = 800.9987 \times 0.5$

$= \mathbf{400.4993\ V}$

(*b*) p.d across non-inductive resistor $= R \times I = 20 \times 0.5 = \mathbf{10\ V}$

(*c*) p.d. across capacitor

$$= X_c \times I = \frac{1}{2\pi f C} \times I = \frac{1 \times 0.5}{2\pi \times 397.887 \times 0.5 \times 10^{-6}} = \mathbf{400\ volts}$$

(*ii*) (*a*) Resistance of circuit = **60 Ω**

(*b*) Inductance of circuit = **0.32 H**

(*iii*) Power Absorbed $= I^2R = (0.5)^2 \times 60 = \mathbf{15\ W}$

Example 5.6. *A coil of resistance 40 Ω and inductance 0.75 H form part of a series circuit for which the resonant frequency is 55 Hz. If the supply voltage is 250 V, 50 Hz, find the following (i) The line current, (ii) The power factor, (iii) The voltage across the coil.*

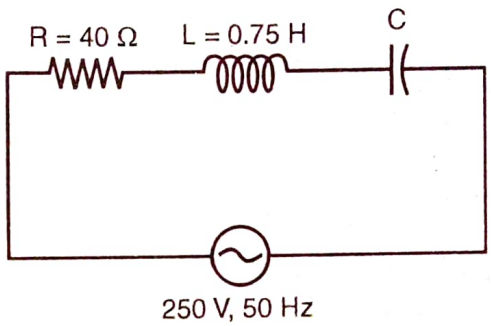


Fig. 5.18.

Solution. Resonant frequency, $f_0 = 55$ Hz

Let, capacitance = C farad

Resonant frequency, $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$

$$C = \frac{1}{(2\pi)^2 f_0^2 L} = \frac{1}{(2\pi)^2 (55)^2 \times 0.75} = 1.1165 \times 10^{-5} = 11.165\ \mu F$$

Voltage applied $= 250\ V,\ 50$ Hz

Reactance of coil $X_L = 2\pi f L = 2\pi \times 50 \times 0.75 = 235.6194\ \Omega$

and Reactance of capacitor, $X_C = \dfrac{1}{2\pi fC} = \dfrac{1}{2\pi \times 50 \times 11.165 \times 10^{-6}} = 285.096\ \Omega$

Total impedance of circuit $= \sqrt{(40)^2 + (235.6194 - 285.96)^2} = 63.6234\ \Omega$

(*i*) Line current, $I = \dfrac{V}{Z} = \dfrac{250}{63.6234} = \mathbf{3.9294\ A}$

(*ii*) Power factor $\dfrac{R}{Z} = \dfrac{40}{63.6234} = \mathbf{0.6287\ (leading)}$

Power factor is leading since capacitive reactance is more than inductive reactance.

(*iii*) Voltage across coil $= I\,X_L = 3.9294 \times 235.6194 = \mathbf{925.8429\ V}$

Example 5.8. *A coil of 15 Ω resistance and 0.3 H inductance is connected in parallel with a variable capacitor across a 230 V, 50 Hz supply. Calculate (i) The capacitance for the resonance (ii) The effective impedence of circuit and (iii) The current supplied from the mains.*

(MACT, Bhopal, 2001)

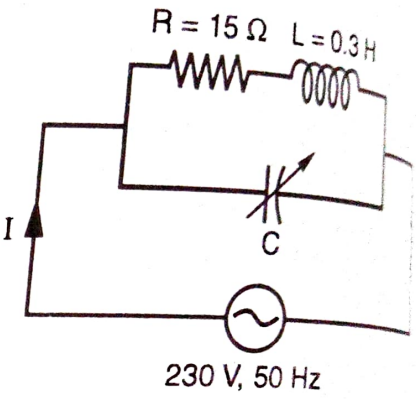


Fig. 5.20.

Solution. Resonant frequency $f_0 = 50$ Hz

Inductance, $L = 0.3$ H Resistance, $R = 15\ \Omega$

At parallel resonance $\dfrac{L}{C} = R^2 + X_L^2$

$\therefore$ $\dfrac{0.3}{C} = (15)^2 + (2\pi \times 50 \times 0.3)^2$

(*i*) Capacitance $C = 3.29 \times 10^{-5}\text{F} = \mathbf{32.93\ \mu F}$

(*ii*) Effective impedance of circuit $= \dfrac{L}{RC} = \dfrac{0.3}{15 \times 3.29 \times 10^{-5}} = \mathbf{607.32\ \Omega}$

(*iii*) Current supplied from mains $= \dfrac{V}{L/RC} = \dfrac{230}{1000} = \mathbf{0.378\ A}$